

Table 1: Major players in wireless mesh networks and their service coverage areas

	Make and model	Coverage area in square metres	Speed	Number of clients/devices supported	Number of units (base+satellites)	Price in rupees (approximate)
1	Linksys WHW0303	557.4	802.11ac (AC3000)	64	1+2	68,000
2	Google Wi-Fi system	418.06	802.11ac (AC1200)	64	1+2	48,000
3	Netgear Orbi Pro AC3000 tri-band Wi-Fi system (SRK60)	464.5	802.11ac (AC3000)	40	1+2	39,000
4	Netgear Orbi RBK50	418.06	802.11ac (AC3000)	40	1+1	25,000
5	Tenda Nova MW6	557.4	802.11ac (AC1200)	90	1+2	23,000
6	Samsung Connect Home AC1300 smart Wi-Fi system	139.3	802.11ac	20	1	20,000
7	TP-Link Deco M5	418.06	802.11ac (AC1300)	64	1+2	17,000

Leading players like Google, Linksys and Tenda Nova have created user friendly interfaces so that you can easily manage and configure your wireless network with the help of a smartphone app. Imagine the impact of extendable routers based on mesh wireless technology, when everything starting from your watch to the TV is a smart and Internet-connected device. In fact, mesh networks can offer uninterrupted Internet connectivity to all your smart devices.

How mesh routers provide seamless connectivity

In mesh topology, your modem first connects to the primary base station and then connects to the satellite stations in weak coverage areas. Let's assume that your primary base station is on the first floor and you have a satellite on the third floor. Now, if you are loading a Web page whose data is being retrieved from the primary base station installed on the first floor and passing this data to the satellite on the third floor, enabling you to view the Web page on that floor, you are actually using the hop process.

Your smartphone, too, will automatically get connected to the strongest Wi-Fi station depending on the location. Hence, this topology not only extends your Wi-Fi range but also helps you to connect to the strongest station. You will be in a fully automated environment where your only job will be to switch on the Wi-Fi option on your smart devices and the mesh Wi-Fi will do its job to provide you with seamless connectivity. **END** 

All about mesh networking wireless technology and how it works

The mesh portal access point generates a radio signal with the wireless mesh service's SSID, a procedure known as black haul. It finds a receiving point signal from another mesh access point that has the strongest RSSI value, establishing a strong connection to the mesh portal SSID.

After the connection is established, it starts providing the services as configured in the controller. If the remote access point fails to connect to the nearest mesh access point within 60 seconds, it tries to connect to the next access point, following a contiguous process of choosing the access points throughout the mesh network to establish a secure connection. If all the access points fail to connect, the main controller reboots and tries again to reconnect by choosing each of the access points.

Planning a mesh topology network is important to get better results. It is recommended that you set the 802.11a radio to mesh services and 802.11g radio to client services. You can configure within the range of 1.09km or less. Also, minimising the mesh access points connected to the mesh portal is a good way to improve the performance of the network topology. Four access points to one mesh portal is the best practice for enhanced wireless connectivity.

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